

Lab 1

NO late submission. Grade = 0 after the deadline.

Due: 13:50h 01/10/2025

- Submission:
 - **Lab#_YOURNAME.ipynb**
 - Must include all data file (**.csv**,...) and **.ipynb** in your submission
 - **.csv** and **.ipynb** must be in the **same** folder (pd.read_csv('data.csv'))
 - No use of LOCAL libraries, NO local directory for libraries

Use '**Lab1_dataset.csv**'.

- a) Missing data could be anything in { NA, n/a, " ", -1, ...}. Read the data and assign the missing data in the dataframe as **NaN**. Hint: df.**replace**(..., np.nan, inplace=True)
- b) Show the summary of the dataframe
- c) Count the **number** and **percentage** of missing data in each column and the entire dataset
- d) **Randomly** drop 25% of the rows that have missing data. Hint: use **any()** and **numpy.random.choice()**
- e) Fill in the missing data with **median** (if the column data type is numeric) or **mode** (if the column data type is not numeric)
- f) Show the summary of the **updated** dataframe
- g) Extract the row with **max** SalePrice and the row with **min** SalePrice
- h) Insert a new column called **Overall**, which is the average of OverallQual and OverallCond.
- i) Update the dataframe with SalePrice used as **index** and **ID** becomes a new column